Assignment 8: Data Analytics with AWS S3, AWS Glue, and Amazon Athena

Overview

This assignment provides hands-on experience with the core AWS serverless data analytics stack.

This is a group assignment. All can practice this in a group.

You will learn to build a modern data lake pipeline using a "bronze-silver-gold" architecture.

You will use **Amazon S3** for scalable data storage, **AWS Glue** to automatically catalog your raw data, and **Amazon Athena** to run interactive SQL queries for transformation and analysis. This process allows you to perform complex data analytics and ETL (Extract, Transform, Load) without provisioning or managing any servers.

Finally, you will analyze real-world weather data, answer business-level questions, and reflect on the significant cost and performance benefits of using serverless tools, partitioning, and columnar formats like Parquet.

Deliverables

Each student must submit the following:

1. **Presentation Slides**
    - Title slide with student name(s), course, and assignment title.
    - Give clear explanation of each completed task.
    - Screenshots or videos of key steps/results.
    - Lessons learned and challenges encountered.

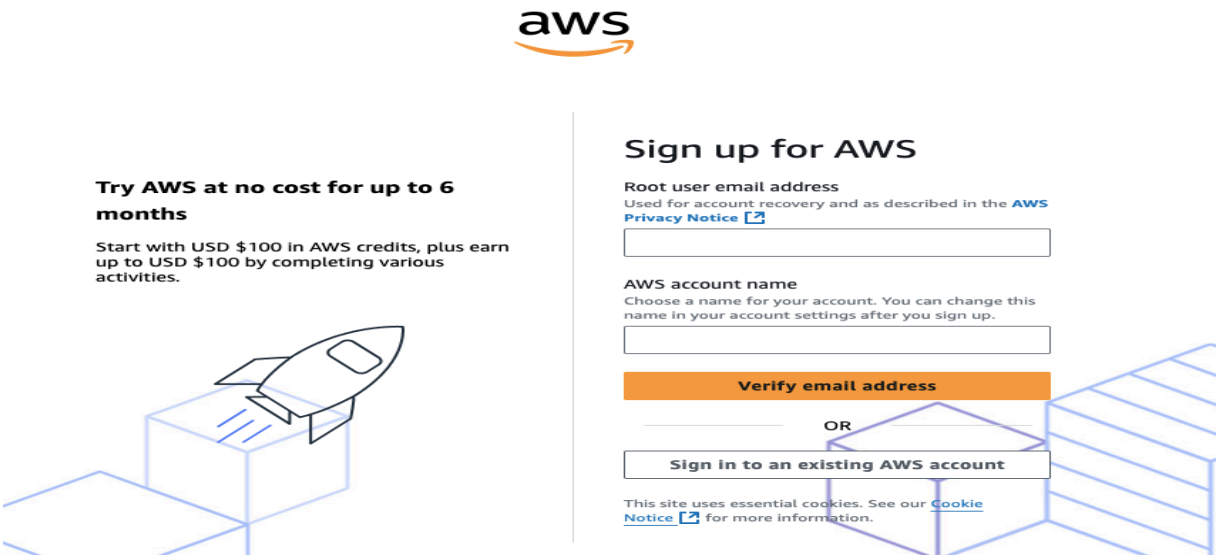
Presentation Guidelines

- Use professional formatting (legible fonts, consistent colors, organized layout).
- Ensure clarity: avoid overcrowding slides with text; use bullet points and visuals.
- Provide screenshots or videos as evidence of work completed.
- Keep presentations within the time limit (8–11 minutes).
- Be prepared to answer questions about the process and challenges.

1. Task Completion (60 points total)
Each AWS task is evaluated based on correctness, completeness, and evidence (e.g., screenshots/videos output).

Open AWS console. https://signin.aws.amazon.com/signup?request_type=register

Register for an AWS account if not created.



2. Presentation Quality (30 points total)

- Clear structure (intro, results, discussion, conclusion) **(10 pts)**
- Slides visually organized and professional, for example, adding pictures, transitions, animations **(10 pts)**
- Screenshots or videos included as evidence **(5 pts)**
- Proper timing (within 6–9 min) **(5 pts)**

3. Professionalism & Delivery (10 points total)

- Confident and clear presentation delivery **(4 pts)**
- Ability to answer questions from peers/instructor **(4 pts)**
- Team collaboration and equal participation **(2 pts)**

Submission Instructions

- Submit slides (PowerPoint or PDF) to Moodle by the deadline.

Task 1: Select and Acquire Raw Data

Learn to identify, understand, and ingest raw source data.

Instructions:

1. Familiarize yourself with the NCEI (NOAA) Global Hourly dataset. This is a common source for public weather data.
   https://www.ncei.noaa.gov/data/global-hourly/

2. Review the CSV_HELP.pdf file (often found with the data) to understand the meaning of key columns like TMP (temperature) and WND (wind).
   https://www.ncei.noaa.gov/data/global-hourly/doc/CSV_HELP.pdf

3. Choose one data file to analyze. For this assignment, we will use a pre-selected file to ensure consistency.

4. Download the following sample file (Washington Dulles, 2019) to your local machine:
   https://www.ncei.noaa.gov/data/global-hourly/access/2019/72403093738.csv

Task 2: Set Up S3 Data Lake Foundation

Establish the "Bronze-Silver-Gold" storage layers in S3. This pattern is an industry best practice for data lakes.

- **Bronze:** Raw, unmodified data, just as it was ingested.
- **Silver:** Cleaned, transformed, and query-optimized data (e.g., in Parquet format).
- **Gold:** Aggregated data, ready for business intelligence and reporting.

Instructions:

1. In the AWS Console, navigate to **Amazon S3**.
2. Create a new, globally unique S3 bucket.
   - **Recommended name:** isd-<YOUR_group_name> (e.g., isd-group11)
3. Inside your new bucket, create three folders (which act as prefixes):
   - bronze/
   - silver/
   - gold/
4. Upload the .csv file you downloaded in Task 1 into the bronze/ folder.

Task 3: AWS Glue Catalog

Use AWS Glue to automatically discover the schema of your raw data and make it queryable with SQL.

Instructions:

1. In the AWS Console, navigate to **AWS Glue**.
2. In the left navigation pane, select **Crawlers**.
3. Click **Create crawler**.
4. **Crawler name:** isd-bronze-crawler
5. **Custom classifiers → Add new classifier**

→ Add a CSV Classifier using default settings Open CSV SerDe with name isd-bronze-classifier.

6. **Data source:** Point the crawler to your S3 bronze/ folder.
    - **Data source path:** s3://isd-<YOUR_GROUP>/bronze/
7. **IAM Role:** Allow Glue to create a new service role for you.
8. **Database:** Create a new database to store your tables.
    - **Database name:** isd_db
9. Finish creating the crawler, then **Run** it.
10. Wait for the crawler to complete (this may take 1-2 minutes). It will report that it has created one table.
11. Navigate to **Amazon Athena** in the AWS Console.
12. In the query editor, you should now see isd_db in the **Database** dropdown and a new table (likely named bronze) in the **Tables** list.
13. Run a test query to confirm it works:

```
SELECT * FROM "isd_db"."bronze" LIMIT 10;
```

Task 4: Transform Data from Bronze to Silver (CTAS)

Convert the raw string data into a clean, typed, and efficient format. You will use a **CREATE TABLE AS SELECT (CTAS)** query in Athena to:

- Parse complex string fields (like TMP and WND).
- Cast data to correct types (e.g., INTEGER, TIMESTAMP).
- Filter out summary rows (e.g., SOD, SOM).
- Save the results as **Parquet**, a columnar format that is much faster and cheaper to query.
- **Partition** the data by year and month for efficient filtering.

Instructions:

1. In the Athena query editor, make sure your isd_db is selected.
2. Run the following CTAS query. **You must modify** the external_location to match your S3 bucket and the FROM line to match your bronze table name.

```
CREATE TABLE "AwsDataCatalog"."isd_db"."silver"
WITH (
  format = 'PARQUET',
  write_compression = 'SNAPPY',
  external_location = 's3://<YOUR_S3_BUCKET>/silver/', -- <-- MODIFY THIS
```

```
  partitioned_by = ARRAY['year','month']
) AS
SELECT
  station,
  CAST(from_iso8601_timestamp(date) AS timestamp) AS ts,

  -- Temperature (°C) from TMP "+0078,1"
  CASE WHEN tmp IS NOT NULL AND substr(tmp,1,5) <> '+9999'
       THEN CAST(substr(tmp,1,5) AS integer)/10.0 END AS temp_c,

  -- Wind speed (m/s) from WND "170,1,N,0015,1"
  CASE WHEN wnd IS NOT NULL AND split(wnd, ',')[4] <> '9999'
       THEN CAST(split(wnd, ',')[4] AS integer)/10.0 END AS wind_ms,

  report_type,
  year(from_iso8601_timestamp(date)) AS year,
  month(from_iso8601_timestamp(date)) AS month

FROM "AwsDataCatalog"."isd_db"."bronze" -- <-- MODIFY THIS if your table name is
different
WHERE report_type NOT LIKE 'SOD%' AND report_type NOT LIKE 'SOM%';
```

3. After the query succeeds, go to your S3 bucket's silver/ folder. You will see new folders for year=2019/ and month=1/, containing Parquet files.

Task 5: Create Aggregated Gold-Layer Views

Create a final "gold" view that aggregates the clean data into daily metrics. This view is what a dashboard or business report would use.

Instructions:

1. In the Athena query editor, run the following CREATE VIEW query. It builds upon your new isd_silver table.

```
CREATE OR REPLACE VIEW "AwsDataCatalog"."isd_db"."gold_daily" AS
SELECT
  date(ts) AS day_utc,
  count_if(temp_c IS NOT NULL) AS obs_count,
  MIN(temp_c) AS t_min_c,
  MAX(temp_c) AS t_max_c,
  AVG(temp_c) AS t_avg_c,
```

```
  AVG(wind_ms) AS wind_avg_ms
FROM "isd_db"."silver"
GROUP BY 1;
```

2. Test your new view to see the daily aggregated weather:

```
SELECT * FROM "AwsDataCatalog"."isd_db"."gold_daily" ORDER BY day_utc;
```

Task 6: Answer Analytics Questions

Use your new data pipeline to answer questions. For each question, provide your SQL query and a brief answer.

Instructions:

1. Run queries against your silver and gold_daily tables to find the answers.
2. **Question 1:** What were the 3 warmest and 3 coldest days (based on max/min temp) in the dataset?
3. **Question 2:** What is the average temperature by the hour of the day (diurnal profile)?
4. **Question 3:** What is the average wind speed by hour? When is it windiest?
5. **Question 4:** Are there any days with incomplete data (e.g., fewer than 24 hourly observations)?

Sample Queries (Adapt these):

```
-- Q1: Warmest / coldest days
SELECT * FROM "isd_db"."gold_daily" ORDER BY t_max_c DESC LIMIT 3;
SELECT * FROM "isd_db"."gold_daily" ORDER BY t_min_c ASC LIMIT 3;

-- Q2: Diurnal profile (avg temp by hour)
SELECT hour(ts) AS hour_utc, AVG(temp_c) AS avg_temp_c
FROM "isd_db"."silver"
GROUP BY 1 ORDER BY 1;

-- Q4: Data completeness
SELECT * FROM "isd_db"."gold_daily" WHERE obs_count < 24 ORDER BY day_utc;
```

Task 7: Cost, Performance, and Cleanup

Reflect on the cost-saving benefits of this architecture and clean up your resources to avoid charges.

Instructions:

1. **Reflection:** In the Athena console, look at the **Recent queries** tab. Find your queries from Task 3 (querying bronze) and Task 6 (querying silver / gold_daily).
    - Note the **"Data scanned"** (e.g., 10 MB) for the query on the raw bronze (CSV) table.
    - Note the **"Data scanned"** (e.g., 1.5 MB) for an equivalent query on the silver (Parquet) table.
    - **Answer:** Why did the query on Parquet scan significantly less data? How does this (and partitioning) directly relate to query speed and cost in Athena?
2. **Cleanup (Avoid Charges):**
    - Navigate to **Amazon S3**.
    - **Empty** your S3 bucket (isd-<YOUR_GROUP>) by deleting all files and folders inside it.
    - **Delete** the S3 bucket itself.
    - Navigate to **AWS Glue**.
    - Delete the **table** (bronze) created by your crawler.
    - Delete the **database** (isd_db).
    - Delete the **crawler** (isd-bronze-crawler).
    - (Athena views/tables are just metadata in the Glue Catalog, so deleting the database removes them).